

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2024 P1H Worked Solutions

Nov 2024 | P1H

31 questions ■ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION**Q#: 1****Marks: 5****TOPIC: Statistics Toolkit**

Nov 2024 P1H - Nov 2024 | P1H

- 1 The table shows some information about the hourly rates of pay of 60 workers.

Hourly rate of pay (p dollars)	Frequency
$10 < p \leq 15$	18
$15 < p \leq 20$	16
$20 < p \leq 25$	14
$25 < p \leq 30$	8
$30 < p \leq 35$	4

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean hourly rate of pay of the 60 workers.

..... dollars
(4)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 5

TOPIC: **Statistics Toolkit**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The modal class is the

The modal class is the class with the greatest frequency:

$$10 < p \leq 15$$

2. Use midpoints

Use midpoints 12.5, 17.5, 22.5, 27.5, 32.5.

$$12.5(18) + 17.5(16) + 22.5(14) + 27.5(8) + 32.5(4) = 1170$$

$$\text{estimated mean} = \frac{1170}{60} = 19.5$$

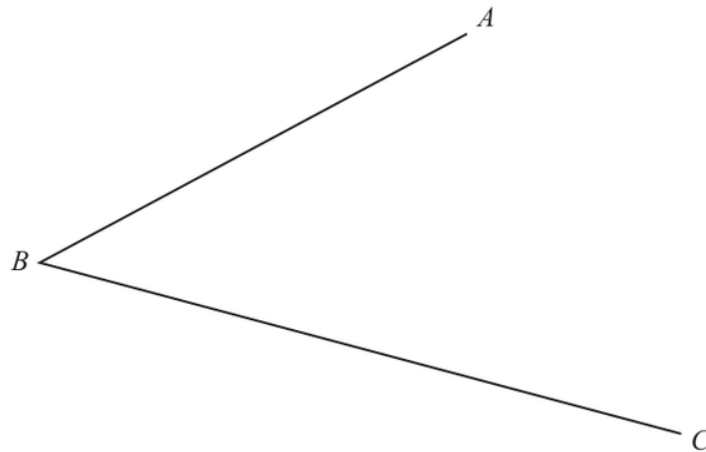
FINAL ANSWER

modal class $10 < p \leq 15$, estimated mean 19.5 dollars.

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2024 P1H - Nov 2024 | P1H

- 2 Use ruler and compasses only to construct the bisector of angle ABC
You must show all your construction lines.



(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Construction answer

To construct the bisector of angle ABC :

1. Draw an arc centred at B cutting both arms of the angle.
2. From the two cut points, draw equal arcs inside the angle.
3. Draw the line from B through the intersection of these arcs.

FINAL ANSWER

The constructed ray is the angle bisector of ABC .

EA

PAST PAPER QUESTION**Q#: 3****Marks: 6****TOPIC: Solving Linear Equations**

Nov 2024 P1H - Nov 2024 | P1H

3 (a) Simplify $(p^3)^5$
(1)(b) Expand and simplify $2n(4n + 3) + n(n - 4)$
(2)(c) Solve $\frac{2x + 5}{3} = 4 - x$

Show clear algebraic working.

 $x =$
(3)**(Total for Question 3 is 6 marks)**

PAST PAPER SOLUTION

Q#: 3

Marks: 6

TOPIC: **Solving Linear Equations**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a) $(p^3)^5 = p^{15}$.

2. (b)

(b)

$$2n(4n + 3) + n(n - 4) = 8n^2 + 6n + n^2 - 4n = 9n^2 + 2n$$

3. (c)

(c)

$$\frac{2x + 5}{3} = 4 - x$$

$$2x + 5 = 12 - 3x$$

$$5x = 7$$

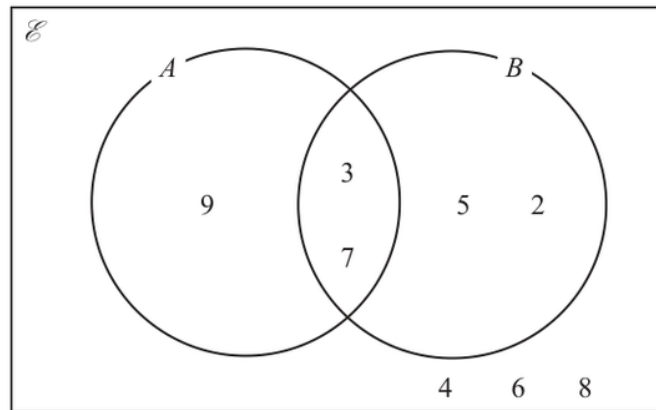
FINAL ANSWER

(a) p^{15} , (b) $9n^2 + 2n$, (c) $x = \frac{7}{5}$

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Nov 2024 P1H - Nov 2024 | P1H

4 Here is a Venn diagram.

(a) List the members of the set B
(1)(b) List the members of the set $A \cap B$
(1)(c) List the members of the set A'
(1)**(Total for Question 4 is 3 marks)**

PAST PAPER SOLUTION

Q#: 4

Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The members of are all

The members of B are all the numbers inside circle B , including the overlap:

$$B = \{2, 3, 5, 7\}$$

2. The members of are the

The members of $A \cap B$ are the numbers in both circles:

$$A \cap B = \{3, 7\}$$

3. Calculate statistic

A' means all the numbers in the universal set that are not in A :

$$A' = \{2, 4, 5, 6, 8\}$$

FINAL ANSWER

(a) $\{2, 3, 5, 7\}$, (b) $\{3, 7\}$, (c) $\{2, 4, 5, 6, 8\}$

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

- 5 The diagram shows a cylinder.

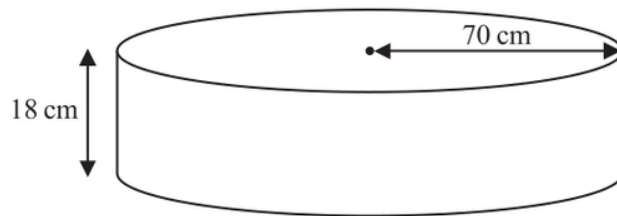


Diagram **NOT**
accurately drawn

The radius of the cylinder is 70 cm

The height of the cylinder is 18 cm

Work out the volume of the cylinder.

Give your answer in litres correct to the nearest litre.

..... litres

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

$$V = \pi r^2 h = \pi (70)^2 (18)$$

$$V = 277088.472 \dots \text{ cm}^3$$

2. Calculate areaSince $1000 \text{ cm}^3 = 1 \text{ litre}$,

$$277088.472 \dots \text{ cm}^3 = 277.088 \dots \text{ litres}$$

FINAL ANSWER

277 litres.

PAST PAPER QUESTION**Q#: 6****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Nov 2024 P1H - Nov 2024 | P1H

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$$B = 2^2 \times 5^2 \times 7^2$$

$$C = 2^2 \times 5^3 \times 7^4$$

Find the highest common factor (HCF) of A , B and C

Write your answer as a product of prime factors.

(Total for Question 6 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the LCM**

$$A = 2^3 \times 5^4 \times 7 \times 11$$

$$B = 2^2 \times 5^2 \times 7^2$$

$$C = 2^2 \times 5^3 \times 7^4$$

2. Find the LCM

The HCF of A , B , and C uses the smallest power of each prime common to all three.

$$2^2, \quad 5^2, \quad 7$$

3. Find the LCM

The prime 11 is not in all three numbers, so it is not included.

$$\text{HCF} = 2^2 \times 5^2 \times 7$$

FINAL ANSWER

$$2^2 \times 5^2 \times 7$$

PAST PAPER QUESTION**Q#: 7****Marks: 4****TOPIC: Percentages**

Nov 2024 P1H - Nov 2024 | P1H

- 7 Shop **A** and Shop **B** have offers for buying the same type of laptop.

The normal price of the laptop in Shop **A** is different to the normal price of the laptop in Shop **B**

Shop A	Shop B
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 4****TOPIC: Percentages**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Shop A discount**

Shop A discount:

$$16\% \text{ of } 475 = 0.16 \times 475 = 76$$

2. Calculate percentage

Shop B: 408 euros is 85% of the normal price.

$$\text{normal price} = 408 \div 0.85 = 480$$

3. Shop B discount

Shop B discount:

$$480 - 408 = 72$$

4. Calculate percentage

Since

$$76 > 72$$

5. Shop A gives the greater

Shop A gives the greater discount.

FINAL ANSWER

Shop A gives the greater discount.

PAST PAPER QUESTION**Q#: 8****Marks: 6****TOPIC: Factorising**

Nov 2024 P1H - Nov 2024 | P1H

8 (a) (i) Factorise $x^2 + 5x - 24$
(2)(ii) Hence, solve $x^2 + 5x - 24 = 0$
(1)(b) Solve the inequality $3y + 5 > 7y - 10$

Show clear algebraic working.

.....
(3)

(Total for Question 8 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 6****TOPIC: Factorising**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)(i)**

(a)(i)

$$x^2 + 5x - 24 = (x + 8)(x - 3)$$

2. Solve quadratic equation(a)(ii) $x = -8$ or $x = 3$.**3. (b)**

(b)

$$3y + 5 > 7y - 10$$

$$15 > 4y$$

$$y < \frac{15}{4}$$

FINAL ANSWER

$$(x + 8)(x - 3), x = -8 \text{ or } x = 3, y < \frac{15}{4}$$

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Nov 2024 P1H - Nov 2024 | P1H

- 9 (a) Write 8.4×10^{-5} as an ordinary number.

.....
(1)

- (b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....
(2)

.....
(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$8.4 \times 10^{-5} = 0.000084$$

2. Convert standard form

For part (b),

$$(6.5 \times 10^{-40})(8 \times 10^{185}) = 52 \times 10^{145}$$

3. Write this in standard form

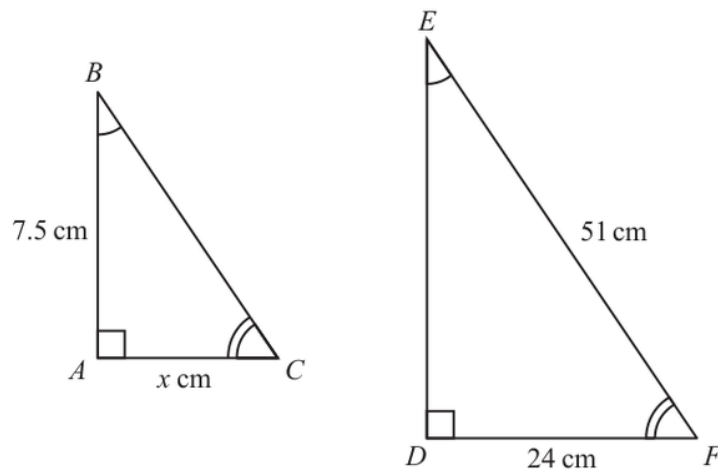
Write this in standard form:

$$52 \times 10^{145} = 5.2 \times 10^{146}$$

FINAL ANSWER(a) 0.000084, (b) 5.2×10^{146}

PAST PAPER QUESTION**Q#: 10****Marks: 5****TOPIC:** Congruence, Similarity & Geometrical Proof

Nov 2024 P1H - Nov 2024 | P1H

10 Here are two similar triangles.Diagrams **NOT**
accurately drawn

Work out the value of x
Show your working clearly.

 $x = \dots\dots\dots$ **(Total for Question 10 is 5 marks)**

PAST PAPER SOLUTION**Q#: 10****Marks: 5****TOPIC: Congruence, Similarity & Geometrical Proof**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

In the larger triangle,

$$DE^2 = 51^2 - 24^2$$

$$DE = 45$$

2. The scale factor from the

The scale factor from the large triangle to the small triangle is

$$\frac{7.5}{45} = \frac{1}{6}$$

3. Evaluate fraction

So

$$x = 24 \times \frac{1}{6} = 4$$

FINAL ANSWER

$$x = 4.$$

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Graphs of Functions**

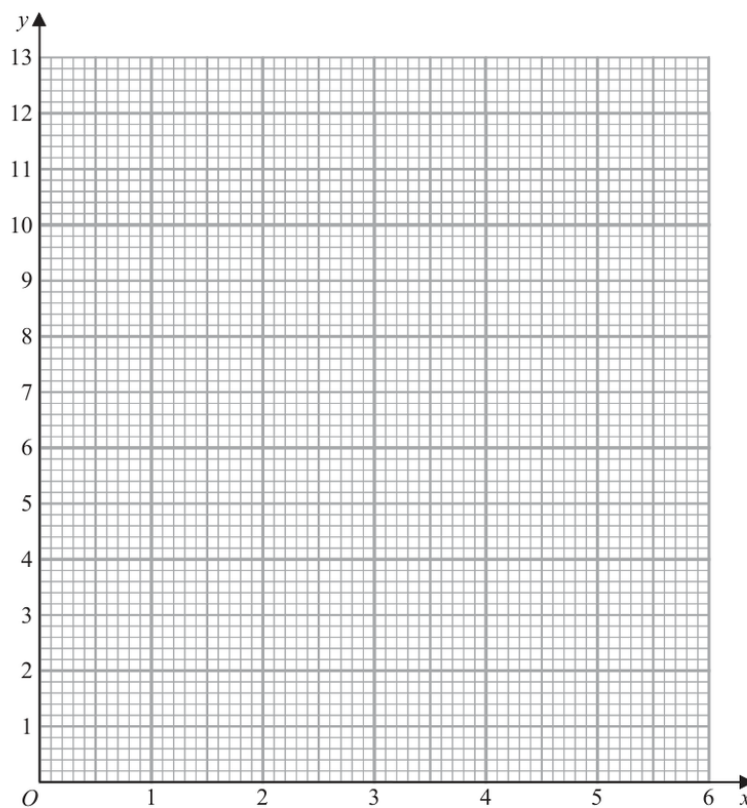
Nov 2024 P1H - Nov 2024 | P1H

- 11 (a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

(2)

- (b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leq x \leq 6$



(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Read the graph**

$$y = 2 \left(x + \frac{1}{x} \right)$$

2. The completed table is

The completed table is:

x	0.5	1	2	3	4	5	6
y	5	4	5	6.7	8.5	10.4	12.3

3. Plot the points and join

Plot the points and join them with a smooth curve.

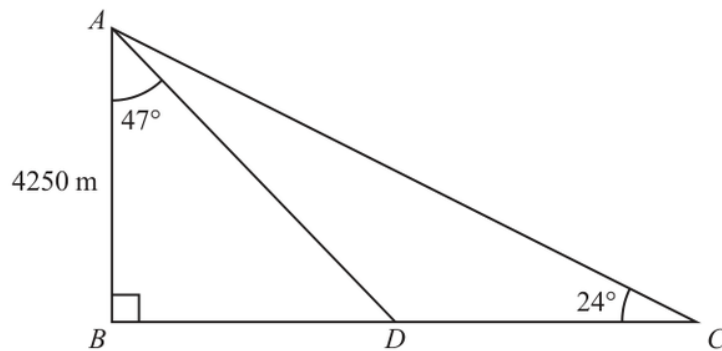
FINAL ANSWER

Missing values are 5, 4, 8.5, 10.4.

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P1H - Nov 2024 | P1H

- 12** ABC is a right-angled triangle.
 D is a point on BC

Diagram **NOT**
accurately drawn

$$AB = 4250 \text{ m} \quad \text{angle } BAD = 47^\circ \quad \text{angle } BCA = 24^\circ$$

Work out the length of DC

Give your answer correct to the nearest integer.

..... m

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**In triangle ABD ,

$$\tan 47^\circ = \frac{BD}{4250}$$

$$BD = 4250 \tan 47^\circ = 4557.567 \dots$$

2. Use trigonometryIn triangle ABC ,

$$\tan 24^\circ = \frac{4250}{BC}$$

$$BC = \frac{4250}{\tan 24^\circ} = 9545.656 \dots$$

$$DC = BC - BD = 4988.089 \dots$$

FINAL ANSWER

4988 m.

PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

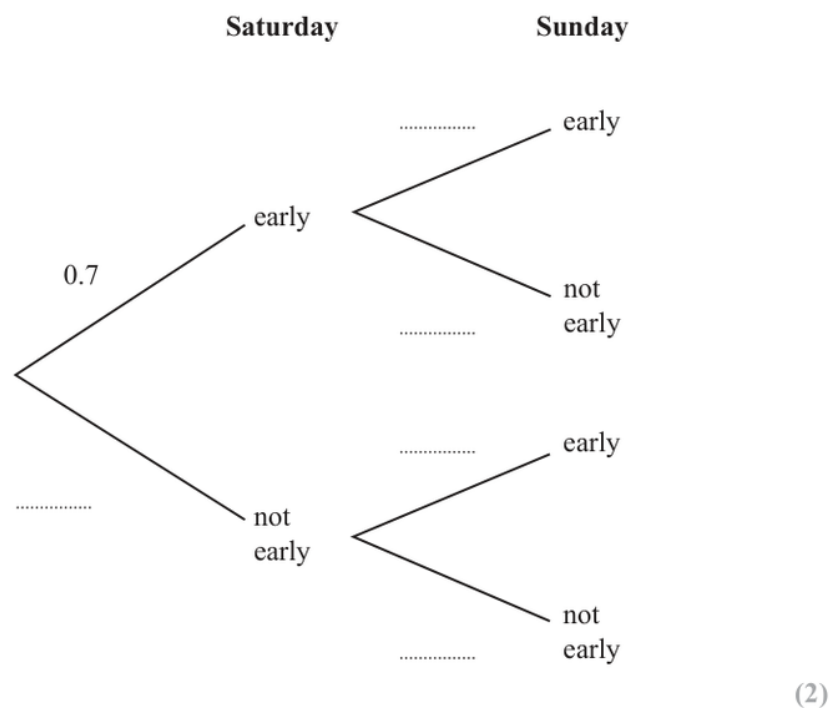
Nov 2024 P1H - Nov 2024 | P1H

13 The probability that Thomas gets to work early on Saturday is 0.7

If Thomas gets to work early on Saturday, the probability that he will get to work early on Sunday is 0.9

If Thomas does **not** get to work early on Saturday, the probability that he will get to work early on Sunday is 0.6

(a) Use this information to complete the probability tree diagram.



(b) Work out the probability that Thomas gets to work early on both Saturday and Sunday.

.....
(2)

(Total for Question 13 is 4 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

$$P(\text{not early on Saturday}) = 1 - 0.7 = 0.3$$

$$P(\text{early on both days}) = 0.7 \times 0.9 = 0.63$$

FINAL ANSWER

0.63.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Direct & Inverse Proportion**

Nov 2024 P1H - Nov 2024 | P1H

14 B is inversely proportional to the square of d

$$B = 0.25 \text{ when } d = 12$$

Find a formula for B in terms of d

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 14****Marks: 3****TOPIC: Direct & Inverse Proportion**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function** B is inversely proportional to d^2 , so

$$B = \frac{k}{d^2}$$

2. Use whenUse $B = 0.25$ when $d = 12$:

$$0.25 = \frac{k}{12^2}$$

$$k = 0.25 \times 144 = 36$$

3. Find inverse function

Therefore

$$B = \frac{36}{d^2}$$

FINAL ANSWER

$$B = \frac{36}{d^2}.$$

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Expanding Brackets**

Nov 2024 P1H - Nov 2024 | P1H

15 Write $3x(2x - 1)(5x + 4)$ in the form $ax^3 + bx^2 + cx$ where a , b and c are integers.

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Expanding Brackets**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Expand brackets**

$$3x(2x - 1)(5x + 4)$$

2. Expand brackets

First expand the two brackets:

$$(2x - 1)(5x + 4) = 10x^2 + 8x - 5x - 4$$

$$= 10x^2 + 3x - 4$$

3. Expand bracketsNow multiply by $3x$:

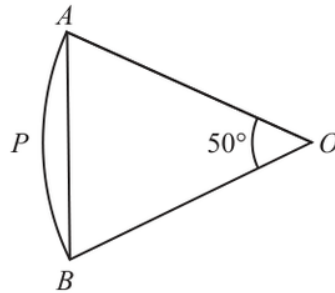
$$3x(10x^2 + 3x - 4) = 30x^3 + 9x^2 - 12x$$

FINAL ANSWER

$$30x^3 + 9x^2 - 12x$$

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P1H - Nov 2024 | P1H

16 $OAPB$ is a sector of a circle, centre O Diagram **NOT**
accurately drawnAngle $AOB = 50^\circ$ Area of triangle $OAB = 120 \text{ cm}^2$ Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2 **(Total for Question 16 is 4 marks)**

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**For triangle OAB ,

$$120 = \frac{1}{2}r^2 \sin 50^\circ$$

$$r^2 = \frac{240}{\sin 50^\circ}$$

2. Area of sectorArea of sector $OAPB$:

$$\frac{50}{360}\pi r^2 = \frac{50}{360}\pi \left(\frac{240}{\sin 50^\circ} \right) = 136.701 \dots$$

FINAL ANSWER137 cm².

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P1H - Nov 2024 | P1H

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

(b) Show that $\frac{5-\sqrt{2}}{\sqrt{2}-1}$ can be written in the form $a+b\sqrt{2}$ where a and b are integers.

(3)

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Algebraic Proof**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\sqrt{675} = \sqrt{25 \times 27} = 5\sqrt{27}$$

2. Calculate valueSo $n = 5$.**3. (b)**

(b)

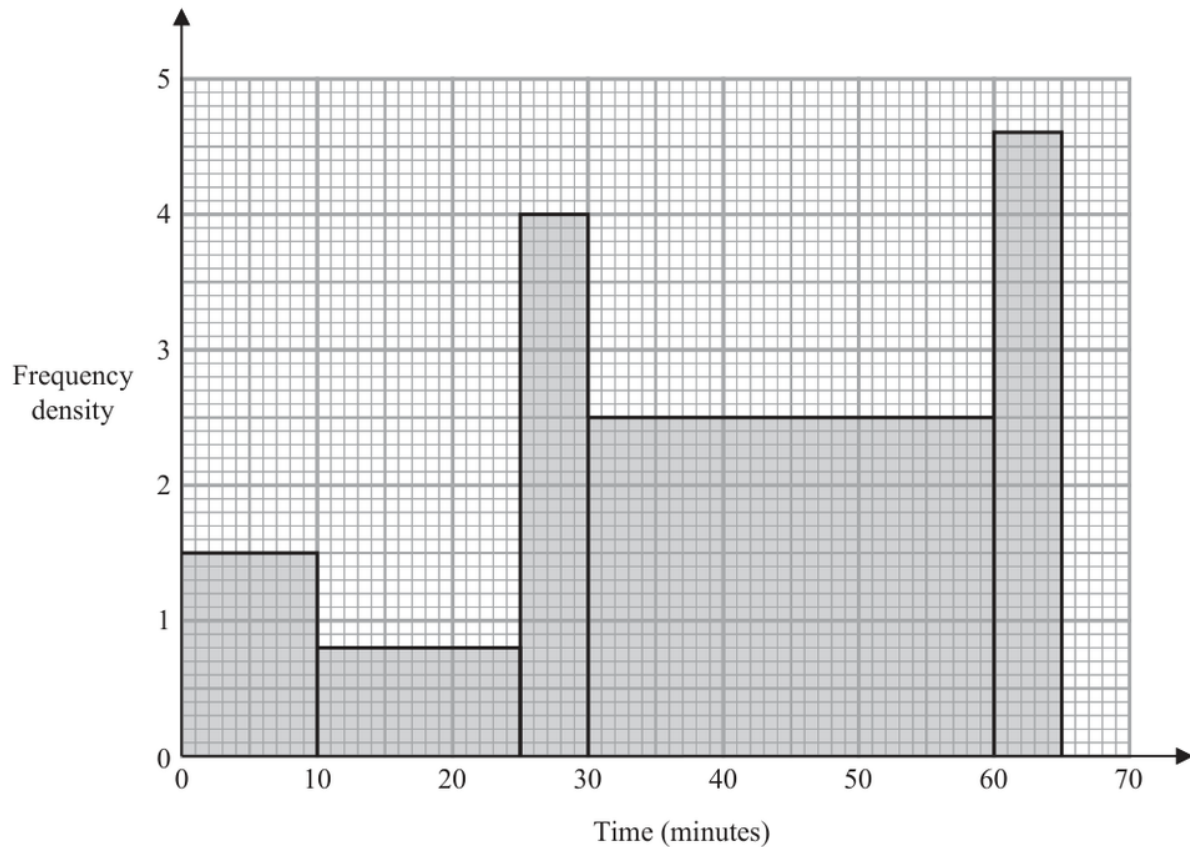
$$\begin{aligned} \frac{5 - \sqrt{2}}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1} &= \frac{(5 - \sqrt{2})(\sqrt{2} + 1)}{2 - 1} \\ &= 5\sqrt{2} + 5 - 2 - \sqrt{2} \\ &= 3 + 4\sqrt{2} \end{aligned}$$

FINAL ANSWER $n = 5$, and $3 + 4\sqrt{2}$.

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Histograms**

Nov 2024 P1H - Nov 2024 | P1H

- 18** The histogram shows information about the times some students took to complete a puzzle.



Work out an estimate for the fraction of these students who took between 20 minutes and 60 minutes to complete the puzzle.

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Histograms**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the frequencies from each**

Estimate the frequencies from each bar:

$$0 \text{ to } 10 : 10(1.5) = 15$$

$$10 \text{ to } 25 : 15(0.8) = 12$$

$$25 \text{ to } 30 : 5(4) = 20$$

$$30 \text{ to } 60 : 30(2.5) = 75$$

$$60 \text{ to } 65 : 5(4.6) = 23$$

2. Total frequency

Total frequency:

$$15 + 12 + 20 + 75 + 23 = 145$$

3. Between and minutes

Between 20 and 60 minutes:

$$5(0.8) + 20 + 75 = 99$$

$$\text{fraction} = \frac{99}{145}$$

FINAL ANSWER

$$\frac{99}{145}$$

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Sequences**

Nov 2024 P1H - Nov 2024 | P1H

19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Sequences**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the first term be**Let the first term be a and the common difference be d .

$$S_{30} = 4395$$

$$\frac{30}{2}(2a + 29d) = 4395$$

$$2a + 29d = 293$$

2. The 8th term plus the

The 8th term plus the 20th term is 284:

$$(a + 7d) + (a + 19d) = 284$$

$$2a + 26d = 284$$

3. Subtract the equations

Subtract the equations:

$$3d = 9$$

$$d = 3$$

4. Find the nth term

Then

$$2a + 26(3) = 284$$

$$2a = 206$$

$$a = 103$$

5. Find the nth termNow find S_{45} :

$$S_{45} = \frac{45}{2}(2a + 44d)$$

$$S_{45} = \frac{45}{2}(206 + 132) = 7605$$

FINAL ANSWER

7605.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

$$y = 2x^4 - 64x$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

3. Find the gradient

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

4. Find the coordinateFind the y -coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

5. The tangent at a minimum

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

FINAL ANSWER

$$y = -96$$

PAST PAPER QUESTION

Q#: 20

Marks: 4

TOPIC: **Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.
Show clear algebraic working.

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Differentiation**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

$$y = 2x^4 - 64x$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

3. Find the gradient

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

4. Find the coordinateFind the y -coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

5. The tangent at a minimum

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

FINAL ANSWER

$$y = -96$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

(Total for Question 21 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$T = \frac{x^2 + y^2}{w}$$

2. Find upper bound

For the upper bound, use the upper bounds of x and y , and the lower bound of w .

$$x < 28.45, \quad y < 17.5, \quad w \geq 87.5$$

$$T_{\text{upper}} = \frac{28.45^2 + 17.5^2}{87.5} = 12.7503\dots$$

3. Estimate the value

Correct to 3 significant figures,

$$12.7503\dots \approx 12.8$$

FINAL ANSWER

12.8

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

(Total for Question 21 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$T = \frac{x^2 + y^2}{w}$$

2. Find upper bound

For the upper bound of T , use the upper bounds of x and y , and the lower bound of w .
 $x = 28.45$, $y = 17.5$, $w = 87.5$

$$T = \frac{28.45^2 + 17.5^2}{87.5}$$

$$T = 12.7503...$$

FINAL ANSWER

12.8 to 3 significant figures.

PAST PAPER QUESTION

Q#: 22

Marks: 3

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

- 22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

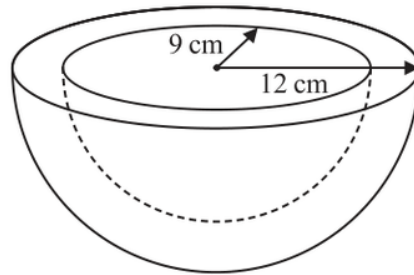


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm

The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm³

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The bowl is a hemispherical**

The bowl is a hemispherical shell.

2. The volume of the metal

The volume of the metal is:

outer hemisphere volume – inner hemisphere volume

3. Volume of a hemisphere

Volume of a hemisphere:

$$\frac{2}{3}\pi r^3$$

4. Calculate volume

So

$$\begin{aligned}\text{metal volume} &= \frac{2}{3}\pi(12)^3 - \frac{2}{3}\pi(9)^3 \\ &= \frac{2}{3}\pi(12^3 - 9^3) \\ &= \frac{2}{3}\pi(1728 - 729) \\ &= 666\pi \\ &= 2092.3007\dots\end{aligned}$$

FINAL ANSWER**2092 cm³**

PAST PAPER QUESTION

Q#: 22

Marks: 3

TOPIC: **Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

- 22 The diagram shows a bowl in the shape of a hemisphere.
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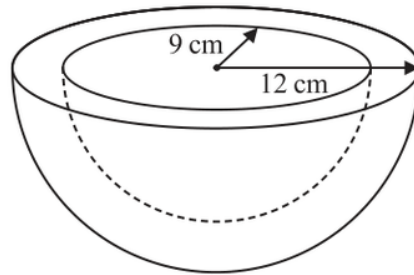


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The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm³

(Total for Question 22 is 3 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Volume & Surface Area**

Nov 2024 P1H - Nov 2024 | P1H

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FINAL ANSWER**2092 cm³**

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

- 23** The curve **C** has equation $y = x^2 - 8x - 9$
 The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)

The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k

Show clear algebraic working.

$k =$

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The curve is**

The curve is

$$y = x^2 - 8x - 9$$

2. The line is

The line is

$$y = k$$

3. Read the graph

At the intersection points,

$$x^2 - 8x - 9 = k$$

$$x^2 - 8x - (9 + k) = 0$$

4. Read the graphThe two roots are p and q .**5. Solve quadratic equation**

For this quadratic,

$$p + q = 8$$

6. We are also given

We are also given

$$p - q = 14$$

7. Solve the two equations

Solve the two equations:

$$2p = 22$$

$$p = 11$$

$$q = -3$$

8. Read the graph

Use one root in $y = x^2 - 8x - 9$:

$$k = 11^2 - 8(11) - 9$$

$$k = 121 - 88 - 9$$

$$k = 24$$

FINAL ANSWER

$$k = 24$$

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

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Given that $p - q = 14$

find the value of k

Show clear algebraic working.

$k =$

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Graphs of Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The curve is**

The curve is

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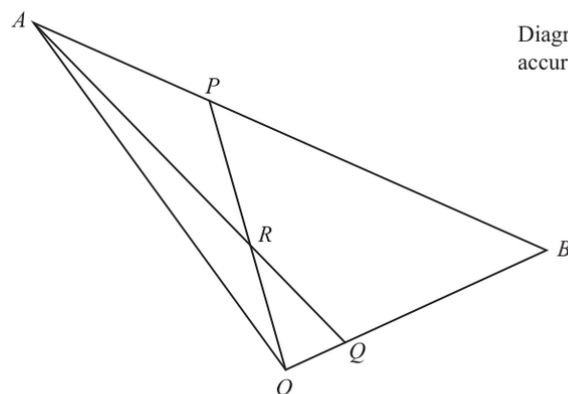
FINAL ANSWER

$$k = 24$$

EA

PAST PAPER QUESTION**Q#: 24****Marks: 6****TOPIC: Vectors**

Nov 2024 P1H - Nov 2024 | P1H

24 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

(iii) $\vec{OR}$

.....
(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION**Q#: 24****Marks: 6**TOPIC: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Read the graph**

$$\vec{OA} = 10\mathbf{a}, \quad \vec{OB} = 10\mathbf{b}$$

2. Read the graph

Also,

$$\vec{AP} = \frac{1}{4}\vec{AB}$$

3. Read the graph

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

4. Read the graph

For part (i),

$$\vec{OQ} = 2\mathbf{b}$$

$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

5. Read the graph

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{OP} = 10\mathbf{a} - \frac{5}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

$$\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

6. Read the graph

For part (iii), R lies on OP , so

$$\vec{OR} = t\vec{OP}$$

7. Read the graph

It also lies on AQ , so

$$\vec{OR} = \vec{OA} + s\vec{AQ}$$

$$\vec{OR} = 10\mathbf{a} + s(2\mathbf{b} - 10\mathbf{a})$$

$$\vec{OR} = (10 - 10s)\mathbf{a} + 2s\mathbf{b}$$

8. Read the graph

Equate with

$$t\left(\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}\right)$$

$$\frac{15}{2}t = 10 - 10s$$

$$\frac{5}{2}t = 2s$$

9. Solve equation

From the second equation,

$$s = \frac{5}{4}t$$

10. Substitute

Substitute:

$$\frac{15}{2}t = 10 - \frac{25}{2}t$$

$$20t = 10$$

$$t = \frac{1}{2}$$

11. Read the graph

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

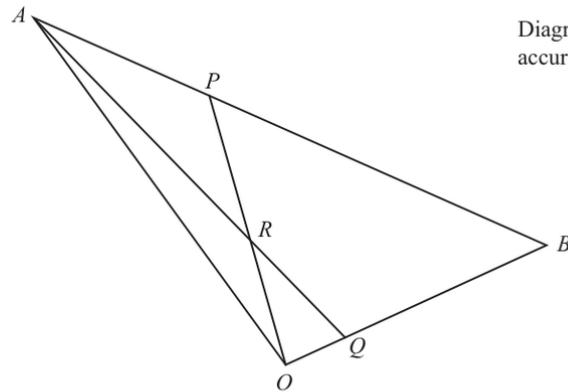
$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

FINAL ANSWER

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}, \vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}, \vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

PAST PAPER QUESTION**Q#: 24****Marks: 6**TOPIC: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

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(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 6

TOPIC: **Vectors**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

$$\vec{OA} = 10\mathbf{a}, \quad \vec{OB} = 10\mathbf{b}$$

2. Read the graph

Also,

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3. Read the graph

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

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$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

5. Read the graph

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

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11. Read the graph

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

FINAL ANSWER

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}, \vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}, \vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

PAST PAPER QUESTION**Q#: 25****Marks: 4****TOPIC: Functions**

Nov 2024 P1H - Nov 2024 | P1H

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 4****TOPIC: Functions**

Nov 2024 P1H - Nov 2024 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = 2x^2 - 24x + 7$$

2. Complete the square

Complete the square:

$$f(x) = 2(x^2 - 12x) + 7$$

$$f(x) = 2[(x - 6)^2 - 36] + 7$$

$$f(x) = 2(x - 6)^2 - 65$$

3. LetLet $y = f(x)$.

$$y = 2(x - 6)^2 - 65$$

$$y + 65 = 2(x - 6)^2$$

$$\frac{y + 65}{2} = (x - 6)^2$$

4. Simplify surdSince $x \geq 6$, take the positive square root:

$$x - 6 = \sqrt{\frac{y + 65}{2}}$$

$$x = 6 + \sqrt{\frac{y + 65}{2}}$$

5. Replace byReplace y by x .**FINAL ANSWER**

$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$

PAST PAPER QUESTION

Q#: 25

Marks: 4

TOPIC: **Functions**

Nov 2024 P1H - Nov 2024 | P1H

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 4****TOPIC: Functions**

Nov 2024 P1H - Nov 2024 | P1H

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$$f(x) = 2x^2 - 24x + 7$$

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$$x = 6 + \sqrt{\frac{y + 65}{2}}$$

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$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$